

Probabilistic Modeling & Optimization

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A general probabilistic ML framework

For a given practical task

- ① **Data:** observed samples $\mathcal{D}$ (e.g., $\{(x_i, y_i)\}_{i=1}^n$ or $\{x_i\}_{i=1}^n$).
- ② **Model:** a parametric probabilistic model $p(\mathcal{D}; \theta)$, which can be $p(\mathbf{X}; \theta)$, $p(\mathbf{Y} \mid \mathbf{X}; \theta)$ or even $p(\mathbf{Y}, \mathbf{X}; \theta)$.
- ③ **Objective:** define a loss/objective $\mathcal{L}(\theta)$ from the model, typically

$$\mathcal{L}(\theta) = -\log p(\mathcal{D}; \theta) \quad (\text{negative log-likelihood}).$$

- ④ **Optimization:** estimate parameters by $\theta^* = \arg \min_{\theta} \mathcal{L}(\theta)$

Setting 1: a basic task

Clinical question:

A patient has a tumor biopsy slide and EHR records.
How severe will the tumor be in the near future?

Data & Formulate the question

Observed dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$$

Features $x_i \in \mathbb{R}^d$

- Cell morphology (size, shape, texture)
- Image-based embedding from tissue sections
- EHR covariates (e.g., age, smoking status, comorbidities)
- RNA sequencing features (gene expression signatures)

Target/Label y_i

- Continuous: growth at 3 months
- Binary: benign vs malignant

Model: single expert regression

Model. Choose a parametric function $e(\cdot; \theta)$ to map features to a prediction:

$$\hat{y}_i = e(x_i; \theta).$$

Examples: linear model, multilayer perceptron (MLP), etc.

Probabilistic assumption (Gaussian noise). Observed outcomes deviate from the prediction due to measurement noise and unmodeled factors:

$$y_i = e(x_i; \theta) + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2).$$

Equivalently,

$$Y_i \mid X_i = x_i; \theta, \sigma^2 \sim \mathcal{N}(e(x_i; \theta), \sigma^2).$$

From model to likelihood

From the noise model,

$$p(y_i | x_i; \theta, \sigma^2) \triangleq \mathcal{N}(y_i; e(x_i; \theta), \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - e(x_i; \theta))^2}{2\sigma^2}\right).$$

Assuming conditional independence across patients:

$$\mathcal{D} \triangleq \{(x_i, y_i)\}_{i=1}^n, \quad \mathbf{x} = (x_1, \dots, x_n), \quad \mathbf{y} = (y_1, \dots, y_n).$$

$$\text{Likelihood: } \ell(\theta) \triangleq p(\mathcal{D}; \theta, \sigma^2) = p(\mathbf{y} | \mathbf{x}; \theta, \sigma^2).$$

$$p(\mathbf{y} | \mathbf{x}; \theta, \sigma^2) = p(y_1, \dots, y_n | x_1, \dots, x_n; \theta, \sigma^2) = \prod_{i=1}^n p(y_i | x_i; \theta, \sigma^2).$$

Negative log-likelihood becomes MSE

Optimization objective: Negative log likelihood (NLL)

$$-\log \ell(\theta) = -\log p(\mathcal{D}; \theta, \sigma^2) = \sum_{i=1}^n \left[\frac{1}{2} \log(2\pi\sigma^2) + \frac{(y_i - e(x_i; \theta))^2}{2\sigma^2} \right]$$

Minimizing NLL is equivalent to:

$$\theta^* = \arg \min_{\theta} \sum_{i=1}^n (y_i - e(x_i; \theta))^2$$

This is the **mean squared error (MSE)** objective (up to scaling).

Key message: MSE is the MLE objective under Gaussian noise.

Optimization: gradient descent / SGD

Define the loss

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - e(x_i; \theta))^2$$

Then gradient descent updates are

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}(\theta)$$

In practice we use mini-batches $\mathcal{B} \subset \{1, \dots, n\}$:

$$\hat{\mathcal{L}}_{\mathcal{B}}(\theta) = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} (y_i - e(x_i; \theta))^2, \quad \theta \leftarrow \theta - \eta \nabla_{\theta} \hat{\mathcal{L}}_{\mathcal{B}}(\theta)$$

(Optional) If the tumor label is binary

If $y_i \in \{0, 1\}$ indicates malignant vs benign, the underlying probabilistic model is:

$$p(Y_i = 1 \mid X_i = x_i; \theta) = \sigma(e(x_i; \theta)), \quad \sigma(t) = \frac{1}{1 + e^{-t}}$$

$$Y_i \mid X_i = x_i; \theta \sim \text{Bernoulli}(\sigma(e(x_i; \theta)))$$

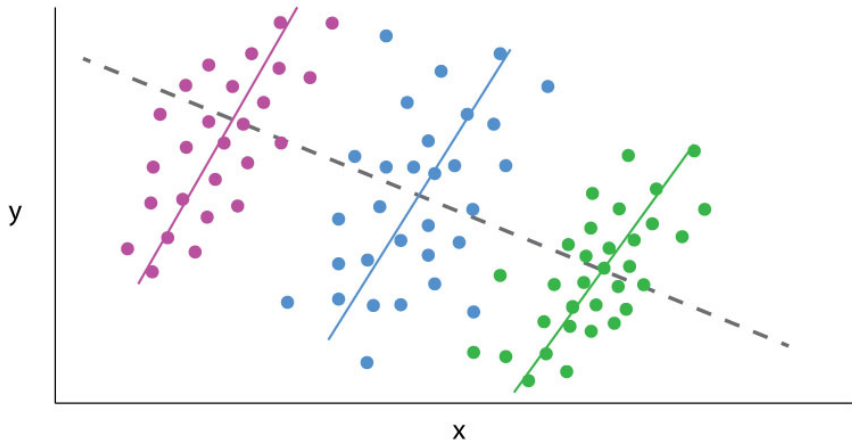
The negative log-likelihood becomes cross-entropy loss:

$$-\log p(\mathcal{D} \mid \theta) = -\sum_{i=1}^n [y_i \log \sigma(e(x_i; \theta)) + (1 - y_i) \log(1 - \sigma(e(x_i; \theta)))]$$

Setting 2: dealing with heterogeneity

Key challenge: tumors are heterogeneous (with different subtypes).

One model can only fit an *average* pattern, which can be misleading (e.g., Simpson's paradox).



Introduce latent subtype variables

Latent subtype:

$$Z_i \in \{1, \dots, K\}, \quad p(Z_i = k \mid X_i = x_i; \phi) = \pi_k(x_i; \phi).$$

Subtype-specific expert (modeled with a parametric function):

$$\mathbb{E}[Y_i \mid X_i = x_i, Z_i = k; \Theta] = e_k(x_i; \theta_k), \quad k = 1, \dots, K.$$

Marginalize the unobserved Z_i (law of total expectation):

$$\begin{aligned} m(x_i) &\triangleq \mathbb{E}[Y_i \mid X_i = x_i; \Theta, \phi] = \mathbb{E}_{Z_i \mid X_i = x_i; \phi} \left[\mathbb{E}[Y_i \mid X_i = x_i, Z_i; \Theta] \right] \\ &= \sum_{k=1}^K p(Z_i = k \mid X_i = x_i; \phi) \mathbb{E}[Y_i \mid X_i = x_i, Z_i = k; \Theta] = \sum_{k=1}^K \pi_k(x_i; \phi) e_k(x_i; \theta_k). \end{aligned}$$

Law of Total Probability & Law of Total Expectation

Law of total probability

$$p(y) = \sum_{k=1}^K p(y \mid Z = k) p(Z = k).$$

Law of total expectation

$$\mathbb{E}[Y] = \mathbb{E}_Z \left[\mathbb{E}[Y \mid Z] \right] = \sum_{k=1}^K \mathbb{E}[Y \mid Z = k] p(Z = k).$$

Same idea: *expand by cases* (condition on Z), then *aggregate* over Z .

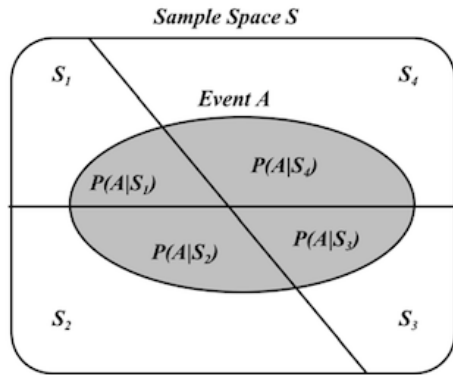
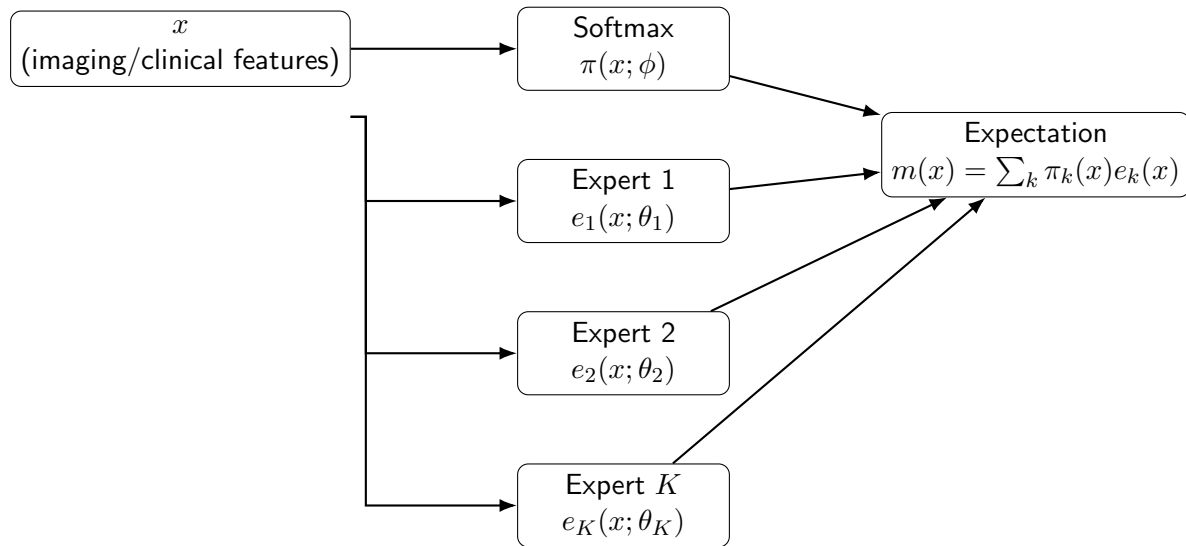


Diagram: mixture of experts (MOE)



Latent model $\Rightarrow$ MSE objective

We then model the outcome with Gaussian noise around the expectation (which goes to **MSE** loss again):

$$Y_i \mid X_i = x_i; \Theta, \phi, \sigma^2 \sim \mathcal{N}(m(x_i), \sigma^2), \quad m(x_i) = \sum_{k=1}^K \pi_k(x_i; \phi) e_k(x_i; \theta_k).$$

Key intuition:

Better predictions come from improving either the experts $e_k(x_i; \theta_k)$ or the assignment weights $\pi_k(x_i; \phi)$ (or both).

Setting 3: when we do not have labels y

Sometimes we only have feature vectors:

$$\mathcal{D}_x = \{x_i\}_{i=1}^n$$

Example: large imaging repository but limited outcomes.

Goal: discover latent tumor subtypes (clusters) directly from feature X .

GMM: probabilistic model with a latent cluster z

Latent cluster variable:

$$Z_i \in \{1, \dots, K\}, \quad p(Z_i = k) = \pi_k.$$

Conditional distribution:

$$X_i \mid Z_i = k \sim \mathcal{N}(\mu_k, \Sigma_k)$$

The marginal density (integrating out z_i):

$$p(x_i) = \sum_{k=1}^K p(Z_i = k) p(x_i \mid Z_i = k) = \sum_{k=1}^K \pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k)$$

$$\log p(\mathcal{D}) = \sum_{i=1}^n \log \left(\sum_{k=1}^K \pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k) \right).$$

Quick contrast: Setting 2 vs GMM (what changes for optimization?)

Setting 2 (supervised): Gaussian around a *marginalized mean*

$$p(y_i | x_i; \Theta, \phi, \sigma^2) = \mathcal{N}(y_i; m(x_i), \sigma^2), \quad -\log p(\mathcal{D}) = -\sum_{i=1}^n \log \mathcal{N}(y_i; m(x_i), \sigma^2).$$

$$\begin{aligned} m(x_i) &\triangleq \mathbb{E}[Y_i | X_i = x_i; \Theta, \phi] = \mathbb{E}_{Z_i | X_i = x_i; \phi} \left[\mathbb{E}[Y_i | X_i = x_i, Z_i; \Theta] \right] \\ &= \sum_{k=1}^K \pi_k(x_i; \phi) e_k(x_i; \theta_k). \end{aligned}$$

GMM (unsupervised): mixture likelihood (latent summed *inside* a log)

$$-\log p(\mathcal{D}) = -\sum_{i=1}^n \log \left(\sum_{k=1}^K \pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k) \right).$$

Takeaway: $\log \sum_k$ structure is **hard for gradient-based optimization**.

Observed data vs. complete data

Observed (incomplete) data: we only see $\mathbf{x} = \{x_i\}_{i=1}^n$.

Latent variables: $\mathbf{z} = \{z_i\}_{i=1}^n$ are unobserved (e.g., cluster IDs).

Complete data: $(\mathbf{x}, \mathbf{z})$. If $\mathbf{z}$ were known, the log-likelihood is often much simpler:

$$\log p(\mathbf{x}, \mathbf{z}; \theta) = \sum_{i=1}^n \log p(x_i, z_i; \theta).$$

But in reality, we must marginalize $\mathbf{z}$:

$$\log p(\mathbf{x}; \theta) = \sum_{i=1}^n \log \sum_{z_i} p(x_i, z_i; \theta),$$

which introduces the $\log \sum(\cdot)$ structure.

E-step: compute posterior responsibilities and take expectation

Observed-data log-likelihood:

$$\ell(\theta) = \sum_{i=1}^n \log p(x_i; \theta) = \sum_{i=1}^n \log \sum_{z_i} p(x_i, z_i; \theta).$$

E-step: compute soft assignments (posterior under current parameters):

$$r_{ik}^{(t)} \triangleq p(Z_i = k \mid X_i = x_i; \theta^{(t)}).$$

Then take an expectation of the complete-data log-likelihood:

$$Q(\theta \mid \theta^{(t)}) \triangleq \mathbb{E}_{\mathbf{Z} \mid \mathbf{X}=\mathbf{x}; \theta^{(t)}} [\log p(\mathbf{x}, \mathbf{z}; \theta)].$$

Interpretation: replace unknown discrete labels z_i by soft labels $r_{ik}^{(t)}$, yielding a deterministic, weighted objective.

(Optional) Why posterior responsibilities & EM efficiency

For any distribution $q_i(z_i)$ with $\sum_{z_i} q_i(z_i) = 1$,

$$\begin{aligned}\log p(x_i; \theta) &= \log \sum_{z_i} q_i(z_i) \frac{p(x_i, z_i; \theta)}{q_i(z_i)} \\ &\geq \sum_{z_i} q_i(z_i) \log \frac{p(x_i, z_i; \theta)}{q_i(z_i)} \quad (\text{Jensen}) \\ &= \sum_{z_i} q_i(z_i) \log p(x_i, z_i; \theta) - \sum_{z_i} q_i(z_i) \log q_i(z_i) \\ &= \mathbb{E}_{Z_i \sim q_i} [\log p(x_i, z_i; \theta)] + \text{const}(q_i).\end{aligned}$$

E-step choice: set $q_i(z_i) = p(z_i \mid x_i; \theta^{(t)})$. This makes the bound *tight* at $\theta = \theta^{(t)}$.

M-step: maximize the bound (equivalently maximize $Q(\theta \mid \theta^{(t)})$), so the observed-data log-likelihood cannot decrease:

$$\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)}).$$

E-step in GMM

GMM complete-data factorization:

$$p(x_i, Z_i = k; \theta) = p(Z_i = k; \boldsymbol{\pi}) p(x_i \mid Z_i = k; \mu_k, \Sigma_k) = \pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k).$$

Choose soft assignments (posterior under current parameters):

$$r_{ik} \triangleq q_i(Z_i = k) = p(Z_i = k \mid X_i = x_i; \theta^{(t)}).$$

Now expand Q :

$$Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^n \mathbb{E}_{Z_i \sim q_i} [\log p(x_i, z_i; \theta)] = \sum_{i=1}^n \sum_{k=1}^K r_{ik} \log \left(\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k) \right).$$

Equivalently,

$$Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^n \sum_{k=1}^K r_{ik} \left[\log \pi_k + \log \mathcal{N}(x_i; \mu_k, \Sigma_k) \right].$$

E-step in GMM: computing responsibilities r_{ik}

Goal: compute the posterior cluster probability (soft assignment)

$$r_{ik} \triangleq p(Z_i = k \mid X_i = x_i; \theta^{(t)}).$$

Bayes rule:

$$p(Z_i = k \mid X_i = x_i; \theta^{(t)}) = \frac{p(x_i \mid Z_i = k; \mu_k^{(t)}, \Sigma_k^{(t)}) p(Z_i = k; \boldsymbol{\pi}^{(t)})}{\sum_{j=1}^K p(x_i \mid Z_i = j; \mu_j^{(t)}, \Sigma_j^{(t)}) p(Z_i = j; \boldsymbol{\pi}^{(t)})}.$$

Plug in GMM components:

$$r_{ik} = \frac{\pi_k^{(t)} \mathcal{N}(x_i; \mu_k^{(t)}, \Sigma_k^{(t)})}{\sum_{j=1}^K \pi_j^{(t)} \mathcal{N}(x_i; \mu_j^{(t)}, \Sigma_j^{(t)})}.$$

Sanity check: $0 \leq r_{ik} \leq 1, \sum_{k=1}^K r_{ik} = 1.$

M-step in GMM: maximize the weighted complete-data objective

M-step: update parameters by maximizing Q :

$$\theta^{(t+1)} = \arg \max_{\theta} Q(\theta \mid \theta^{(t)}).$$

Let $N_k = \sum_{i=1}^n r_{ik}$. The maximizers are:

$$\pi_k^{(t+1)} = \frac{N_k}{n}, \quad \mu_k^{(t+1)} = \frac{1}{N_k} \sum_{i=1}^n r_{ik} x_i,$$

$$\Sigma_k^{(t+1)} = \frac{1}{N_k} \sum_{i=1}^n r_{ik} (x_i - \mu_k^{(t+1)})(x_i - \mu_k^{(t+1)})^\top.$$

Interpretation: treat r_{ik} as fractional labels and perform weighted MLE.

EM vs Gradient Descent (GD)

Same objective:

$$\max_{\pi, \mu, \Sigma} \sum_{i=1}^n \log \left(\sum_{k=1}^K \pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k) \right)$$

Gradient-based optimization:

- Works because the objective is differentiable (log-sum-exp).
- Requires handling constraints: $\pi_k \geq 0$, $\sum_k \pi_k = 1$ and $\Sigma_k \succ 0$.
- Needs step-size/optimizer choices; can be slower for this classical model.

EM optimization:

- Uses responsibilities r_{ik} to build a weighted complete-data problem.
- For GMM, M-step is closed form (fast, no learning rate).
- Typically converges efficiently in practice for standard GMMs.

Summary

Setting	Probabilistic model	Optimization
Supervised, single expert	$Y \mid X = x \sim \mathcal{N}(e(x; \theta), \sigma^2)$	NLL $\Rightarrow$ MSE; GD/SGD
Supervised, MOE	$Y \mid X = x \sim \mathcal{N}(\sum_k \pi_k(x) e_k(x), \sigma^2)$	NLL $\Rightarrow$ MSE on weighted mean; GD/SGD
Unsupervised clustering	$Z \sim \text{Cat}(\boldsymbol{\pi}), X \mid Z = z \sim \mathcal{N}(\mu_z, \Sigma_z)$	Maximize $\sum_i \log \sum_k \pi_k \mathcal{N}(\cdot);$ EM

Takeaway: define $p(\mathcal{D}; \theta)$ first; the loss and optimizer follow.

Appendix: estimating σ^2 in Gaussian regression

If σ^2 needs to be estimated in Setting 1, the NLL is

$$\sum_{i=1}^n \left[\frac{1}{2} \log(2\pi\sigma^2) + \frac{(y_i - e(x_i; \theta))^2}{2\sigma^2} \right]$$

For fixed θ , the MLE for σ^2 is

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - e(x_i; \theta))^2.$$

Appendix: expectation over z in detail

In Setting 2, We can write out the proof of the expectation in detail

$$\begin{aligned} m(x_i) &\triangleq \mathbb{E}[Y_i \mid X_i = x_i; \Theta, \phi] = \int y p(y \mid x_i; \Theta, \phi) dy \\ &\stackrel{(*)}{=} \int y \left(\sum_{k=1}^K p(z_i = k \mid x_i; \phi) p(y \mid x_i, z_i = k; \Theta) \right) dy \\ &= \sum_{k=1}^K p(z_i = k \mid x_i; \phi) \underbrace{\int y p(y \mid x_i, z_i = k; \Theta) dy}_{\mathbb{E}[Y_i \mid X_i = x_i, Z_i = k; \Theta]} \\ &= \sum_{k=1}^K \pi_k(x_i; \phi) \mathbb{E}[Y_i \mid X_i = x_i, Z_i = k; \Theta] = \sum_{k=1}^K \pi_k(x_i; \phi) e_k(x_i; \theta_k). \end{aligned}$$

Appendix: posterior makes the bound tight in EM

$$\begin{aligned}\mathcal{L}(q, \theta) &\triangleq \sum_z q(z) \log \frac{p(x, z; \theta)}{q(z)} \\&= \sum_z q(z) \log p(x, z; \theta) - \sum_z q(z) \log q(z) \\&= \sum_z q(z) (\log p(z | x; \theta) + \log p(x; \theta)) - \sum_z q(z) \log q(z) \\&= \log p(x; \theta) - \sum_z q(z) \log \frac{q(z)}{p(z | x; \theta)} \\&= \log p(x; \theta) - \text{KL}(q(z) \parallel p(z | x; \theta)).\end{aligned}$$